Shift-Share Designs and Event Studies

Topics in Macroeconomics – Week 2

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PhD elective, 2026/27

Bank of Portugal & Católica Lisbon

Today

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Material draws on Borusyak, Hull & Jaravel (2025), Hull (Applied Econometrics, ch. 8) and Borusyak (ARE 213 lectures).

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$\Rightarrow$ The answer to the second depends on the answer to the first

Why shift-share?

The shift-share instrument

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- Where it comes from: the treatment itself has this form, $x_\ell = \sum_n s_{\ell n}\, x_{\ell n} \rightarrow$ replace the local growth rates $x_{\ell n}$ by common ones g_n

[Bartik (1991), Borusyak, Hull & Jaravel (2025)]

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Card (2009)	city	migrants by origin	new migrants by origin
Autor, Dorn & Hanson (2013)	comm. zone	industry employment	Chinese imports elsewhere
Nakamura & Steinsson (2014)	state	procurement sensitivity	national military build-up
Greenstone, Mas & Nguyen (2020)	county	bank market shares	bank lending shocks
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$\Rightarrow$ Most cross-sectional macro evidence is a shift-share design, named or not

[Borusyak, Hull & Jaravel (2025), Table 1]

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- **Findings:** more exposed areas saw higher unemployment, lower participation, lower wages and rising transfer payments; import competition explains about a quarter of the decline in US manufacturing employment
- **Why it matters:** launched the "China shock" literature (labour markets, politics, innovation) and became the textbook shift-share design: GPSS, AKM and BHJ all re-analyse it

[Autor, Dorn & Hanson (2013, 2016), Autor, Dorn, Hanson & Majlesi (2020)]

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- **Ideal experiment:** assign Chinese productivity gains at random across industries; regions differ in how exposed they are through their industry mix

[Autor, Dorn & Hanson (2013), Borusyak, Hull & Jaravel (2025)]

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$$x_{\ell t} = \sum_n \frac{L_{\ell nt}}{L_{\ell t}} \cdot \frac{\Delta M_{nt}^{US}}{L_{nt}}, \qquad z_{\ell t} = \sum_n \underbrace{\frac{L_{\ell n,t-10}}{L_{\ell,t-10}}}_{s_{\ell nt}} \cdot \underbrace{\frac{\Delta M_{nt}^{OTH}}{L_{n,t-10}}}_{g_{nt}}$$

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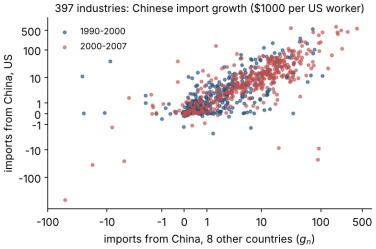
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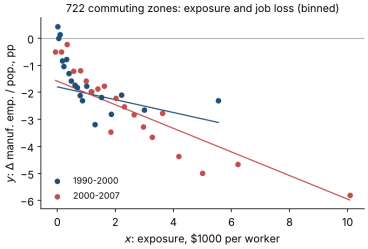
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$\Rightarrow$ A translation device: industry-level shocks, region-level outcomes

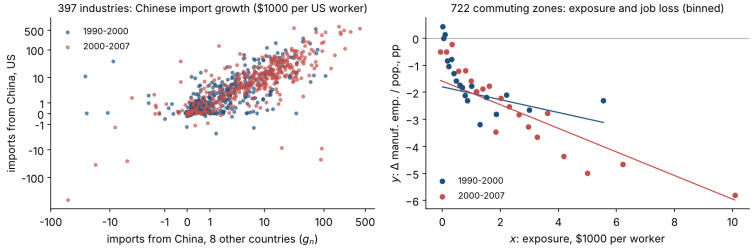
[Autor, Dorn & Hanson (2013), Section I]

The data: shifts and outcomes



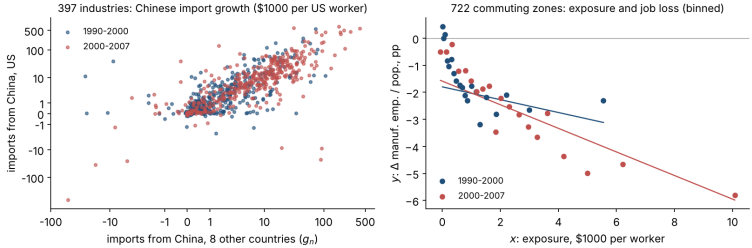


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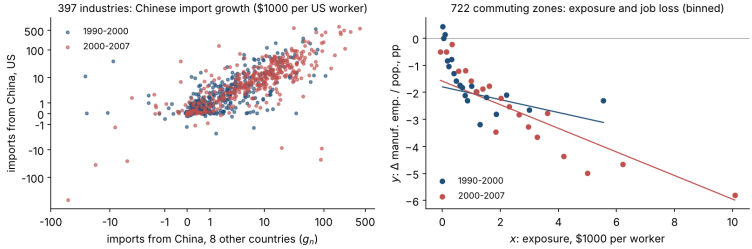
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$\Rightarrow$ Right panel = the simple regression (S1); the instrument keeps only the exposure driven by the left panel

One instrument, two commuting zones

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2000–07	California CZ	Michigan CZ
Manufacturing share, $S_{\ell t} = \sum_n s_{\ell nt}$	0.31	0.34
Largest industries	chips, computers	office furniture, car parts
Their shocks g_{nt}	42, 189	0.6, 8.5
Instrument $z_{\ell t} = \sum_n s_{\ell nt}\, g_{nt}$	10.7	2.9
of which computers alone	4.9	0.0
Exposure $x_{\ell t}$	7.3	3.1
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$\Rightarrow$ The instrument asks: did CZs specialised in the industries where China took off lose more jobs?

[CZs 37500 and 12200; data: Goldsmith-Pinkham, Sorkin & Swift (2020) replication files]

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  S5 The shock-level view S6 Rotemberg weights S7 Balance

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$\Rightarrow$ Every number and figure comes from `week2_simple.py` (one cell per step) and `week2_fast.py`

Two routes to identification

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$\Rightarrow$ Pick one and say which: controls, inference and diagnostics differ

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$\Rightarrow$ ADH and Acemoglu & Restrepo (2020) use the same shares for different treatments: they cannot both be valid share instruments

Route 1: exogenous shares

- The shift-share IV is a weighted average of one-share-at-a-time IV estimates:

$$\hat{\beta} = \sum_k \hat{\alpha}_k \hat{\beta}_k, \qquad \hat{\beta}_k = \frac{s_k' My}{s_k' Mx}, \qquad \hat{\alpha}_k = \frac{g_k \, s_k' Mx}{\sum_j g_j \, s_j' Mx}$$

  - M: residualise on the controls; $\hat{\alpha}_k$ = Rotemberg weights (sum to one, can be negative)
- **Credible when** shares are tailored to the treatment: migrant shares by origin only measure exposure to migration [Card (2009)]
- **Not credible when** shares are generic: industry shares measure exposure to *every* industry shock (technology, demand, robots . . .)

$\Rightarrow$ ADH and Acemoglu & Restrepo (2020) use the same shares for different treatments: they cannot both be valid share instruments

- **Diagnostics:** which shares drive $\hat{\beta}$ (S6), balance on those shares, over-identification

[Goldsmith-Pinkham, Sorkin & Swift (2020)]

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$\Rightarrow$ The natural route for the China shock: many industries, shocks driven by Chinese supply

[Borusyak, Hull & Jaravel (2022)]

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- A constant at the shock level becomes the control $S_{\ell} = \sum_{n} s_{\ell n}$ in the CZ regression (next slide)

[Borusyak, Hull & Jaravel (2022)]

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$\Rightarrow$ Exposure to random shocks is not random when exposure itself is non-random

[Borusyak, Hull & Jaravel (2022)]

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$\Rightarrow$ **The route decides the SEs:** SEs redraw the random part

- Route 2, random shocks: CZs with similar industries move together $\Rightarrow$ exposure-robust
- Route 1, random shares, fixed shocks: CZs are the independent units $\Rightarrow$ conventional

[Adão, Kolesár & Morales (2019), Borusyak, Hull & Jaravel (2022)]

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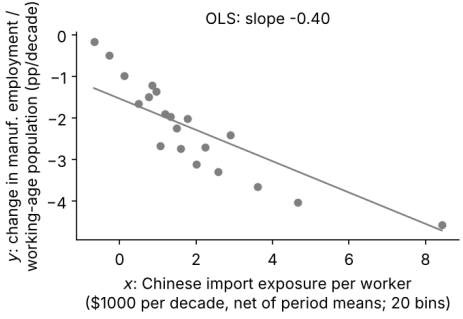
$\Rightarrow$ For shares: tailored shares, Rotemberg weights, balance on top shares, over-identification

[Borusyak, Hull & Jaravel (2025), Table 2]

The China shock, step by step

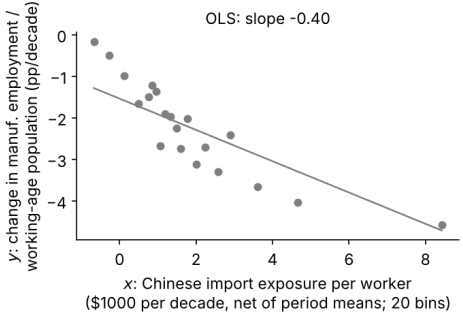
S1: OLS

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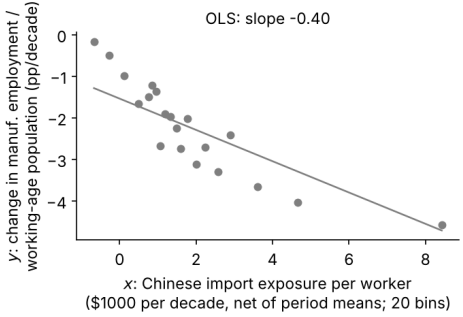
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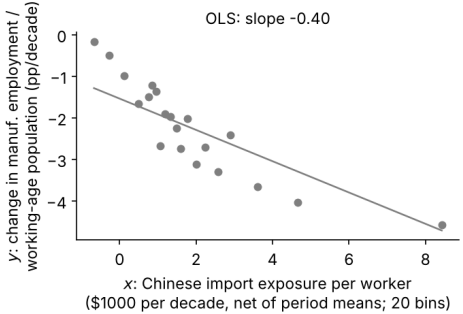
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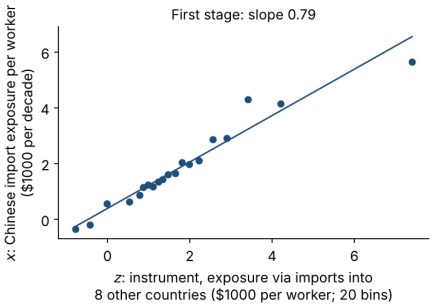


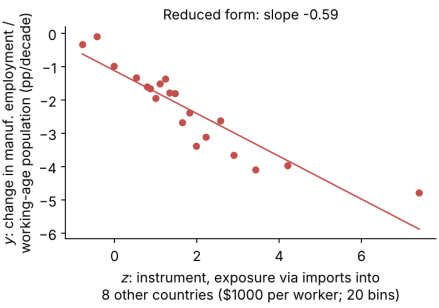
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$\Rightarrow$ Biased towards zero if US demand shocks raise imports *and* jobs

S2: the ADH instrument

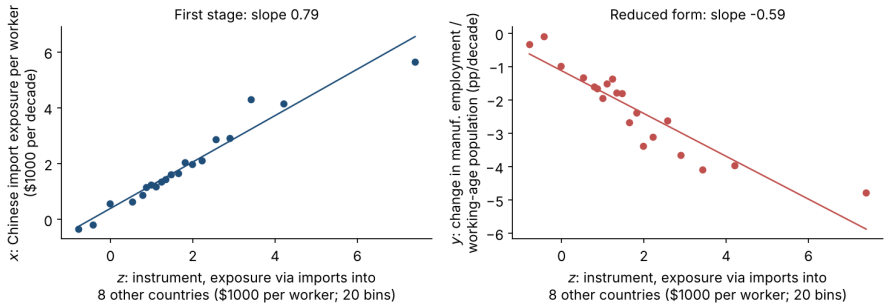
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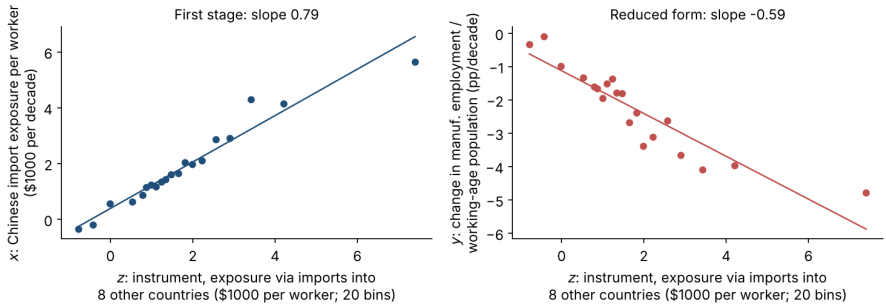
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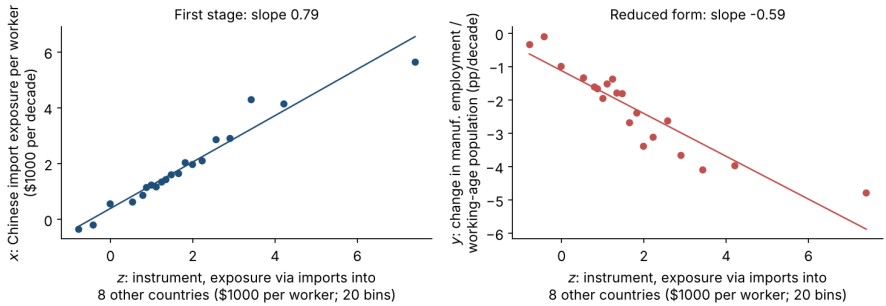
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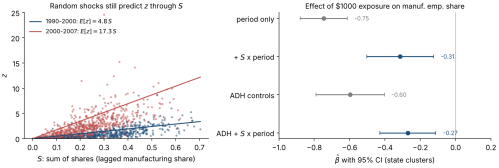


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$\Rightarrow$ IV $>$ OLS in absolute value, as the demand story predicts

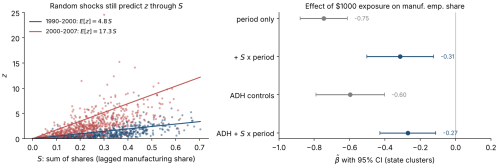
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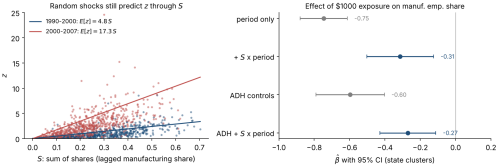
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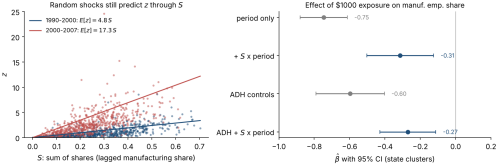
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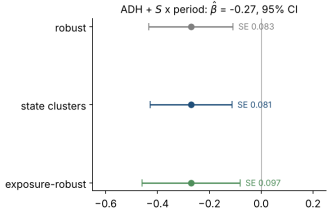
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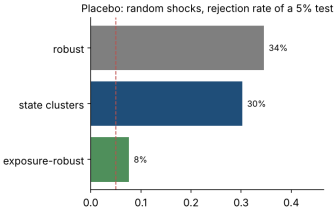
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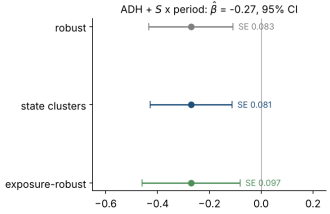
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- Controlling for $S \times$ period: $-0.75 \to -0.31$; with ADH's controls (w): $-0.60 \to -0.27$ (0.08)
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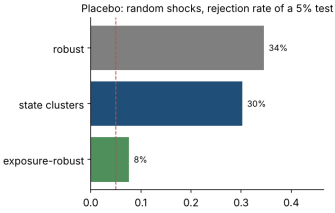
S4: which standard errors?





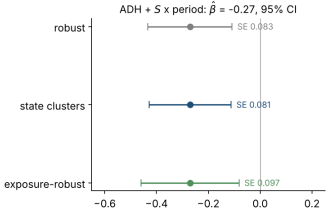
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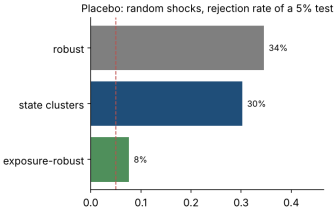




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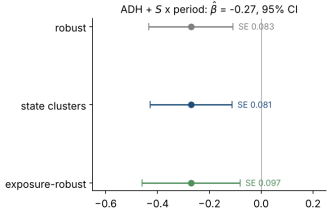
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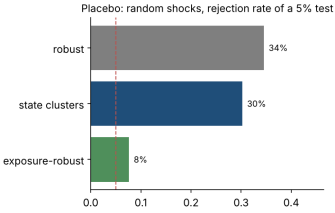




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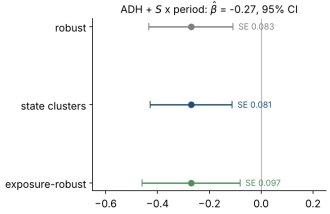
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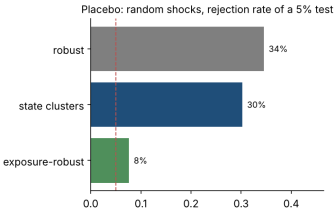




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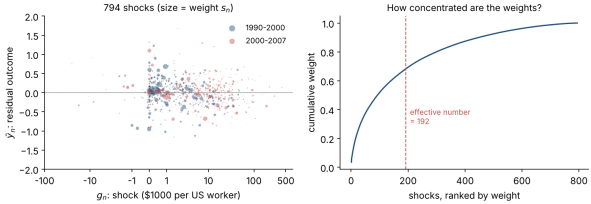




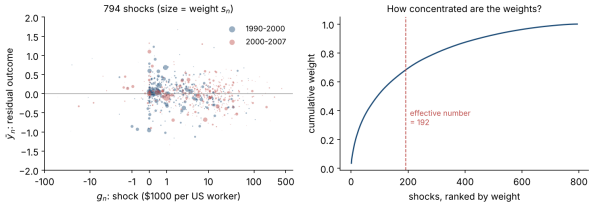
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$\Rightarrow$ Our estimate: SE 0.081 $\rightarrow$ 0.097, still significant. In other designs the correction can kill the result

S5: the shock-level view

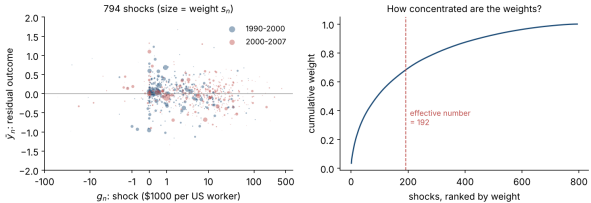


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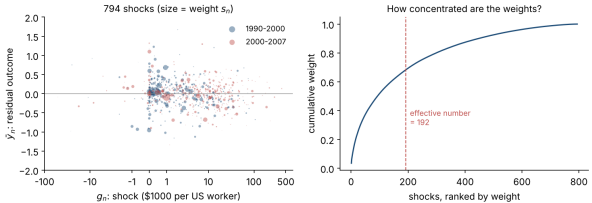
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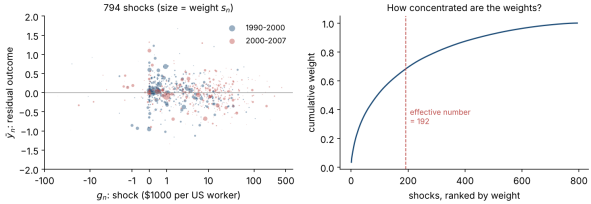
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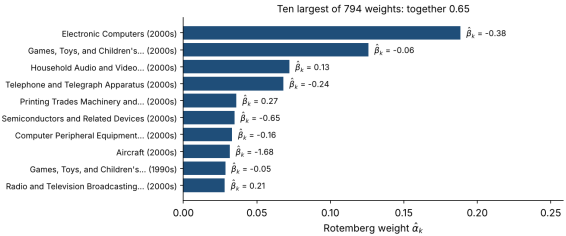
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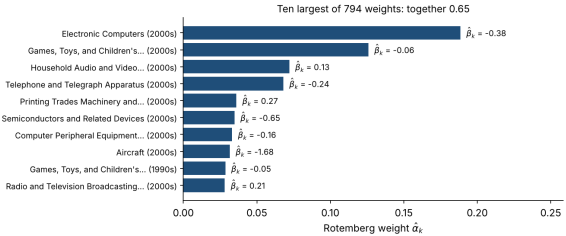
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$\Rightarrow$ Many small shocks, none dominant: A2 is plausible. With a dozen, a few industries would decide $\hat{\beta}$

S6: Rotemberg weights

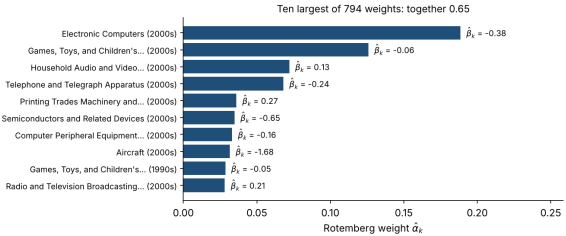


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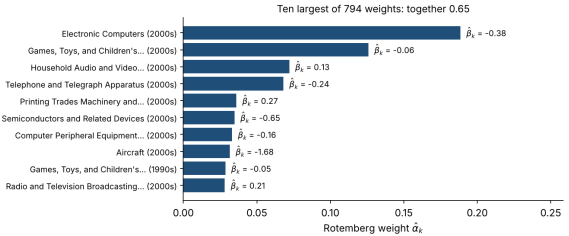
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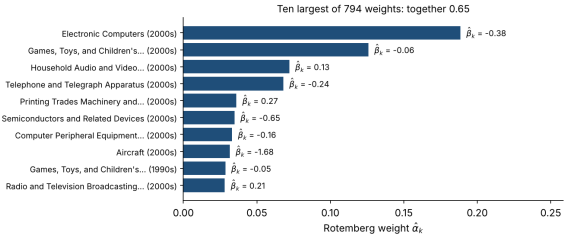
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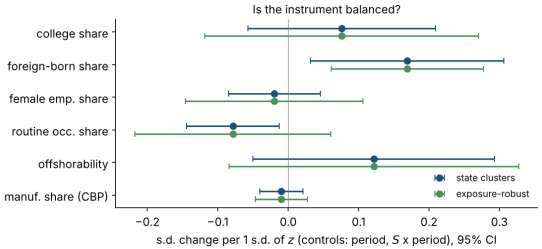


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$\Rightarrow$ Always ask which part of the data drives the number. Check: drop computers $\Rightarrow -0.26$, robust

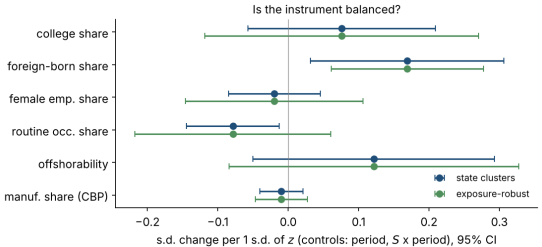
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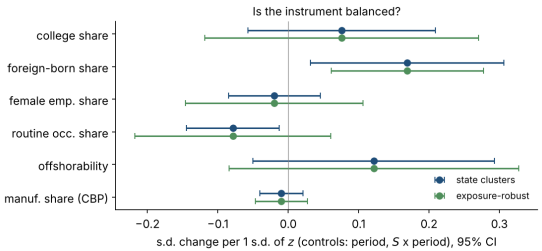
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⇒ A reason to keep ADH's controls; check also shock-level balance (industry skill, wages, capital intensity) and pre-trends

[Borusyak, Hull & Jaravel (2022, 2025)]

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- Ways forward:
  - A model disciplined by the cross-sectional estimate [Nakamura & Steinsson (2014), Adão, Arkolakis & Esposito (2019)]
  - Aggregate time-series evidence for the intercept (Week 1)

Event studies with staggered adoption

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- **Link to shift-share:** same panel logic, units with different exposure, net of common time effects
  - Shift-share: how much a unit is exposed. Staggered: when
  - Same question as S6: *what does the regression actually compare?*
  - Shift-share: set by random shocks
  - Staggered TWFE: set by timing, already-treated units act as controls

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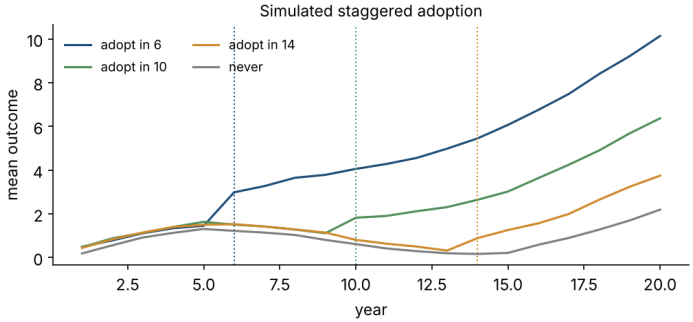
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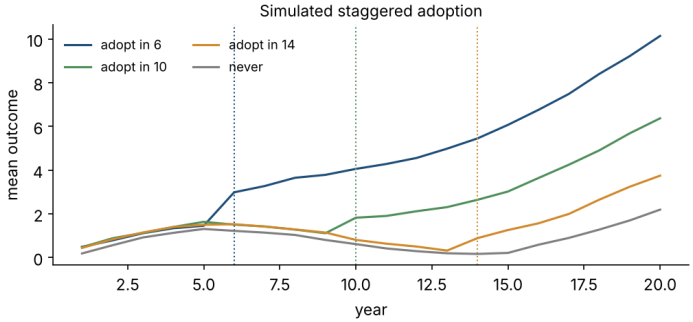
$$\begin{aligned} \text{static TWFE:} \quad & Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + e_{it} \\ \text{event study:} \quad & Y_{it} = \alpha_i + \lambda_t + \textstyle\sum_{k \neq -1} \beta_k \, 1\{t - E_i = k\} + e_{it} \end{aligned}$$

[Borusyak, Jaravel & Spiess (2024)]

E0: a simulated panel where we know the truth

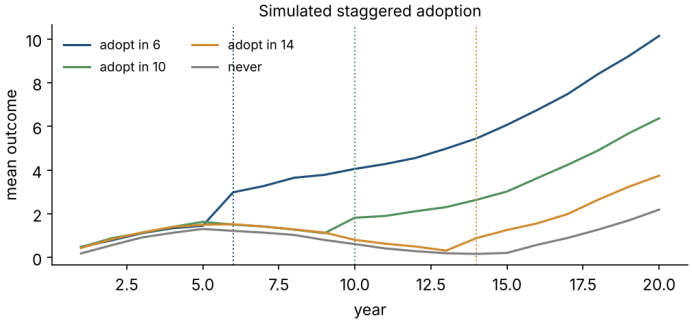


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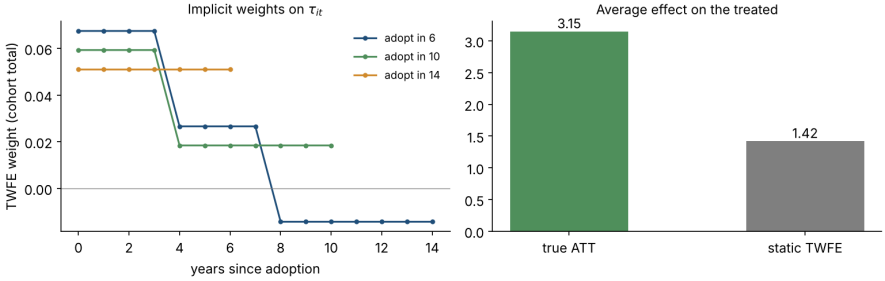
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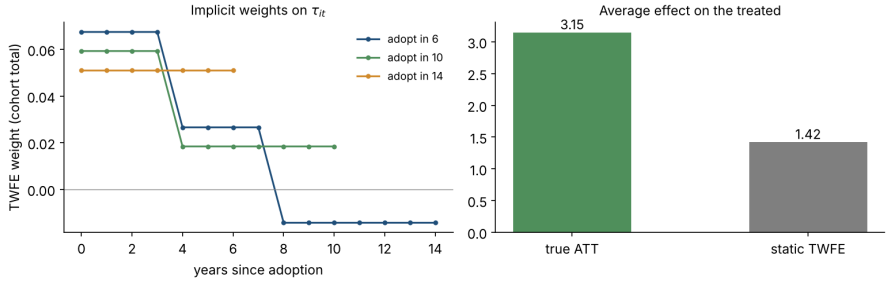
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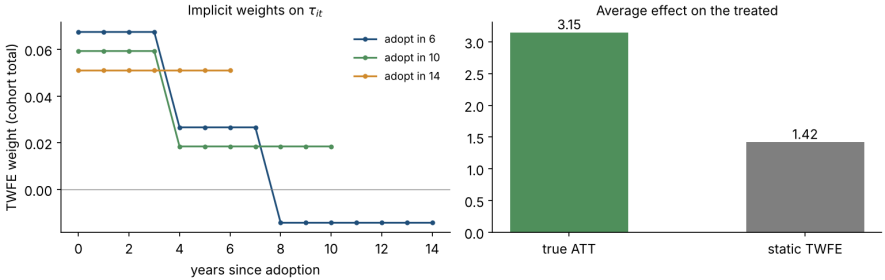


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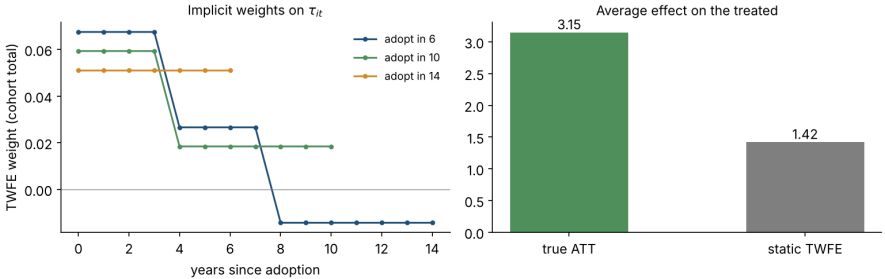
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$\Rightarrow$ TWFE overweights short-run effects and puts negative weight on the largest ones

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- **Fix** [Borusyak, Jaravel & Spiess (2024)]: compare treated cells only with untreated ones – the imputation estimator
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- **Pre-trends:** regress untreated Y_{it} on unit and year effects plus lead dummies – a separate test

[Borusyak, Jaravel & Spiess (2024); also Callaway & Sant'Anna (2021), de Chaisemartin & D'Haultfœuille (2024)]

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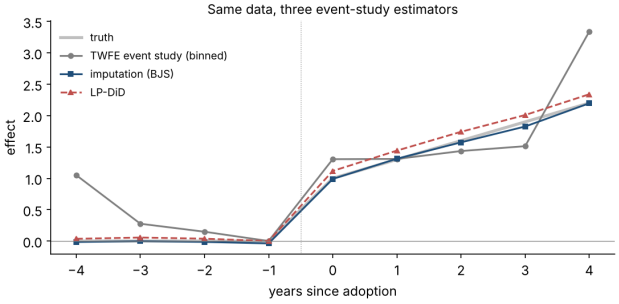
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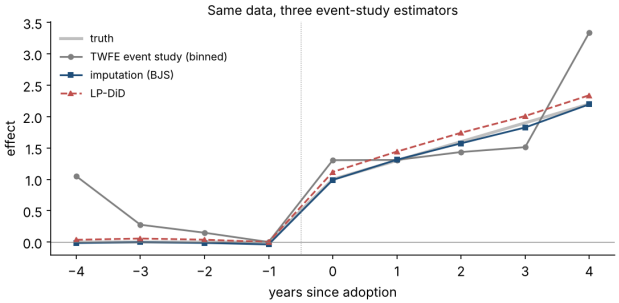
$\Rightarrow$ Clean-control restriction = the identifying assumption, like the controls in an LP

[Dube, Girardi, Jordà & Taylor (2025)]

E2: same data, three event-study estimators

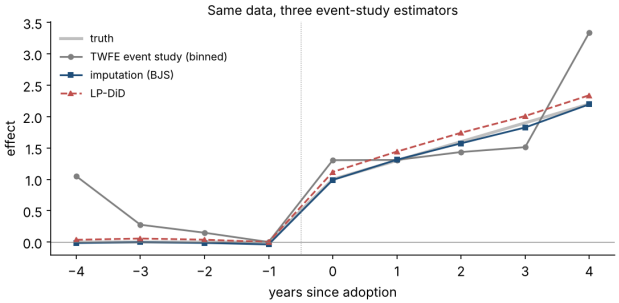


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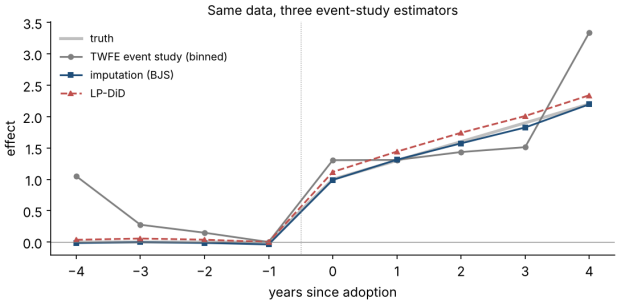
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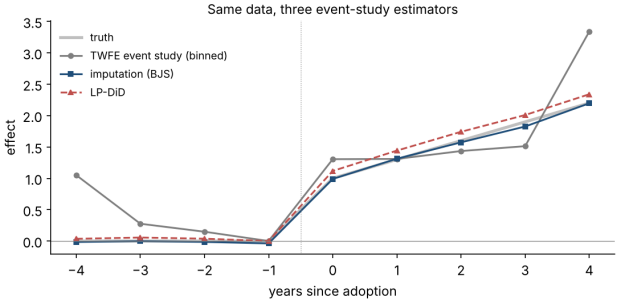
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$\Rightarrow$ Same data, same parallel trends: only the comparisons differ

How it fits together

LPs, VARs and shift-share designs

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$\Rightarrow$ Ask every design: what is random, how many draws of it are there, what does the regression weight?

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- Next: implementation: the same steps in code

Implementation

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- **Exercise** ($\sim$25 minutes): `exercise_ADH_student.py`, six short parts with `check()` lines

The instrument in code: a merge and a group-by

```
shares = pd.read_stata("data/Lshares.dta")            # czone, year, sic87dd, ind_share
shocks = pd.read_stata("data/shocks.dta")             # year, sic87dd, g

m = shares.merge(shocks[["year", "sic87dd", "g"]], on=["year", "sic87dd"])
m["sg"] = m.ind_share * m.g                           # s_ln * g_n
inst = m.groupby(["czone", "year"]).agg(z=("sg", "sum"), S=("ind_share", "sum")).reset_index()
df = df.merge(inst, on=["czone", "year"])
df["S_90"], df["S_00"] = df.S * (1 - df.t2), df.S * df.t2     # S x period

ADH = "l_shind_manuf_cbp + l_sh_popedu_c + ... + C(division)"
res = IV2SLS.from_formula(f"d_sh_empl_mfg ~ 1 + t2 + S_90 + S_00 + {ADH} + [shock ~ z]",
                          df, weights=df.weights).fit(cov_type="clustered", clusters=df.statefip)
print(res.params["shock"])                            # -0.270
```

Exposure-robust SEs in code: the shock-level IV

```
def shock_level(cols, controls, data):
    r = data[["czone", "year", "weights"]].copy()
    for c in cols:                                    # 1. residualise on the CZ controls
        r[c] = smf.wls(f"{c} ~ {controls}", data, weights=data.weights).fit().resid
    sl = shares.merge(r, on=["czone", "year"])        # 2. attach residuals to shares
    sl["ws"] = sl.weights * sl.ind_share
    for c in cols:
        sl[c] = sl[c] * sl.ws
    out = sl.groupby(["year", "sic87dd"])[cols + ["ws"]].sum()     # 3. average by shock
    out[cols] = out[cols].div(out.ws, axis=0)
    out = out.reset_index().merge(shocks[["year", "sic87dd", "g"]], on=["year", "sic87dd"])
    out["sic3"] = (out.sic87dd // 10).astype(int)
    return out

sh = shock_level(["d_sh_empl_mfg", "shock"], PREF, df)              # 4. IV across shocks
r = IV2SLS.from_formula("d_sh_empl_mfg ~ 0 + C(year) + [shock ~ g]", sh, weights=sh.ws) \
          .fit(cov_type="clustered", clusters=sh.sic3)                # -0.270 (0.097)
```

Event studies in code: imputation and LP-DiD

```
# Imputation: unit and year effects from untreated cells only, then average by k
imp = smf.ols("Y ~ C(i) + C(t)", panel[panel.D == 0]).fit()
panel["tau_hat"] = panel.Y - imp.predict(panel)
es_imp = [panel[panel.k == k].tau_hat.mean() for k in range(0, 9)]

# LP-DiD: one long-difference regression per horizon, newly treated vs clean controls
def lpdid(h):
    rows = []
    for t in range(2, T + 1):
        if not 1 <= t + h <= T:
            continue
        newly = (wideD[t] == 1) & (wideD[t - 1] == 0)
        clean = wideD[max(t, t + h)] == 0
        keep = newly | clean
        rows.append(pd.DataFrame({"dy": wideY[t + h][keep] - wideY[t - 1][keep],
                                  "dD": newly[keep].astype(int), "t": t}))
    return smf.ols("dy ~ dD + C(t)", pd.concat(rows)).fit().params["dD"]
```

$\Rightarrow$ Same loop over h as the Week 1 LP; only the sample changes with h

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`exercise_ADH_student.py`: fill in the `# TODO` lines; each `check()` prints ✓ when your number matches.

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- **Take-home** (due Week 3): the BHJ checklist for the China shock; spillovers with exposure-robust SEs; event-study estimators with cohort-specific constant effects

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- **Next week:** financial crises and policy in open economies – sudden stops, collateral constraints, macroprudential policy